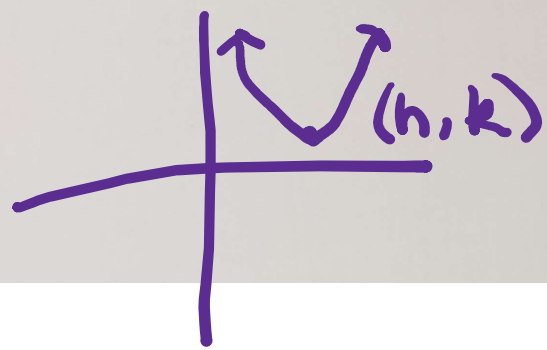
Vertex to polynomial form

$y = a(x-h)^2 + k$ vertex is (h, k)

$y = ax^2 + bx + c$



Vertex form to polynomial form
Where do we start?

$$y = a(x-h)^2 + k$$

Need to undo the exponent first.
It is exponent over addition/subtraction so can't distribute directly.

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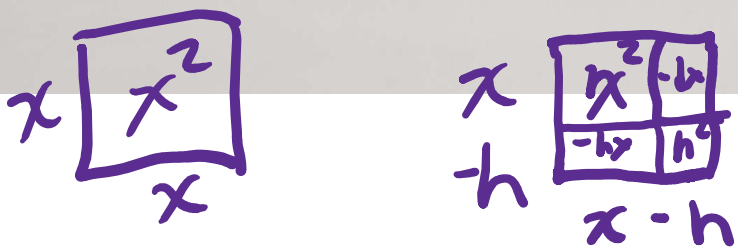


Need to undo the exponent first.
It is exponent over addition/subtraction so can't distribute directly.

$(x-h)^2$ can't distribute

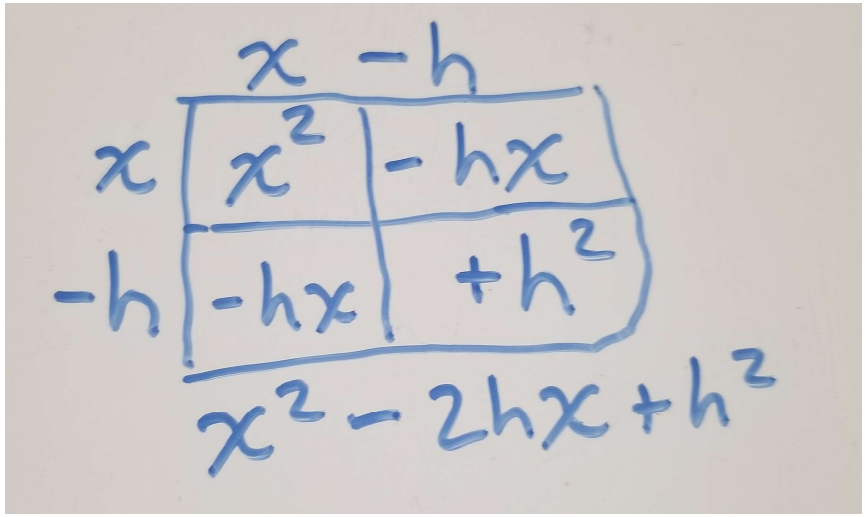
$(x-h)(x-h)$ change to multiplication

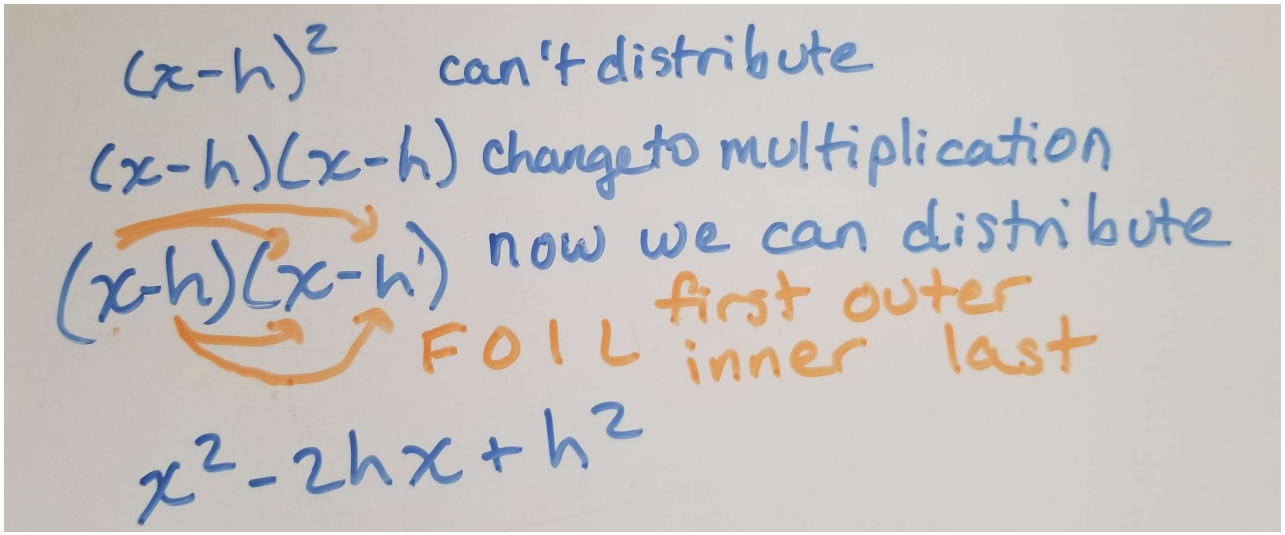
$(x-h)(x-h)$ now we can distribute

FOIL first outer inner last

$x^2-2hx+h^2$

Or can use the box method to distribute instead of FOIL





Now can put the expanded exponent part into the original equation.
What do we do next?

$$y = a(x-h)^2 + k$$

$$y = a(x^2 - 2hx + h^2) + k$$

Next, we distribute out the a by multiplying it by each term of the expanded exponent. Notice that the k is not multiplied by the a.

$$y = a(x-h)^2 + k$$

$$y = a(x^2 - 2hx + h^2) + k$$

Here is the equation written out in expanded form. What are the a, b, and c in the standard form?

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$$y = a(x^2 - 2hx + h^2) + k$$

$$y = ax^2 - 2ahx + ah^2 + k$$

$$y = ax^2 + bx + c$$

Here is the equation written out in expanded form. What are the a, b, and c in the standard form?

$$y = a(x-h)^2 + k$$

$$y = a(x^2 - 2hx + h^2) + k$$

$$y = ax^2 \underbrace{- 2ah}_{b}x + \underbrace{ah^2 + k}_{c}$$

$$y = ax^2 + bx + c$$

$$a = a$$

$$b = -2ah$$

$$c = ah^2 + k$$

Vertex form to polynomial form when a is 1

$$y = (x-h)^2 + k$$

$$(x-h)(x-h) + k$$

$$x^2 - 2hx + h^2 + k$$

$$Ax^2 + Bx + C$$

$$A = 1$$

$$B = -2h$$

$$C = h^2 + k$$

Vertex form to polynomial form

$$(x-h)^2+k$$

$$(x-h)(x-h)+k$$

$$x^2-2hx+h^2+k$$

A B C

Vertex to polynomial form

$$y = a(x-h)^2 + k$$

$$y = a(x^2 - 2hx + h^2) + k$$

$$y = ax^2 \underbrace{-2ah}_{b}x + \underbrace{ah^2 + k}_{c}$$

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